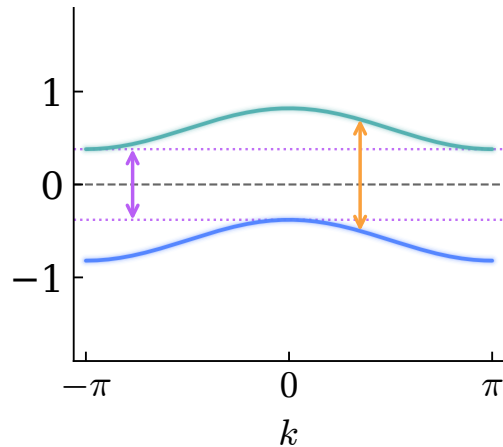


Energy (arbitrary units)

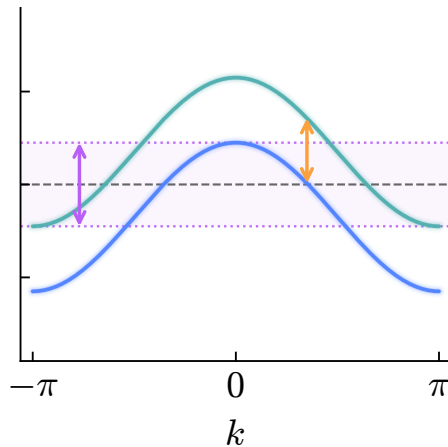
(a) Global gap

$$\Delta_{\text{dir}}^{\text{min}} > 0, \Delta_{\text{ind}} > 0$$



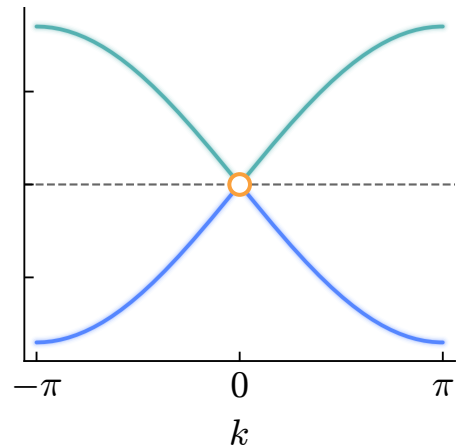
(b) Energy overlap

$$\Delta_{\text{dir}}^{\text{min}} > 0, \Delta_{\text{ind}} < 0$$



(c) Band contact

$$\Delta_{\text{dir}}^{\text{min}} = 0$$



— $E_{2N_p, \vec{k}}$

— $E_{2N_p+1, \vec{k}}$

----- $E_F = 0$

— $\Delta_{\text{dir}}(k)$

— Δ_{ind}

○ Band contact